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Key Points:

- Wet heat waves dominating humid southern China have shown a sharp increase
- Future projections suggest a more pronounced increase in wet heat waves than dry heat waves
- About 40% of people will be exposed to dangerous wet heat waves in a warmer future, 4 times that of dry heat waves

Supporting Information:

Supporting Information may be found in the online version of this article.

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Divergent Risks of Dry and Wet Heat Waves Across China: Historical Trends and Future Projections

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Abstract Heat waves have been intensifying with climate change, presenting a growing risk to human health and ecosystems. However, the differential impacts of dry versus wet heat waves and their future trajectories remain underexplored. Here, we introduce a data-driven approach to classify heat waves by relative humidity, revealing dry heat waves predominantly affect China's arid northwest, while wet heat waves are more prevalent in its humid south. These events' frequency and persistence have increased substantially over recent decades. Under a high socioeconomic scenario, the intensified dry and wet heat waves will continue expanding, with increases of approximately $141\% \pm 48\%$ and $224\% \pm 59\%$ by the 21st century's end. Population exposure to wet heat waves is projected to increase by about $100\% \pm 36\%$, nearly doubling that of dry heat waves, with North China facing the highest risks. By then, an estimated 10% and 40% of China's population will face exposure to dangerous dry and wet heat waves. Adopting a moderate socioeconomic development pathway could substantially mitigate these risks. Our findings indicate that climate change is reshaping these heat wave threats, necessitating targeted adaptation strategies to address these evolving challenges.

Plain Language Summary Heat waves are becoming more frequent and severe due to climate change, but their impacts vary depending on whether they are dry (low humidity) or wet (high humidity). This study examines how these two types of heat waves have affected China in the past and how they might change in the future. Using observational data and model projections, the researchers found that dry heat waves are most common in China's arid northwest, while wet heat waves dominate the humid south. Over the past six decades, both types of heat waves have increased in frequency, with wet heat waves rising more rapidly, especially in southern China. Under high-emission scenarios, wet heat waves could surge by 224% by the end of the century. Nearly 40% of China's population could be exposed to dangerous wet heat waves, with North China facing the most severe risk. Dry heat waves will also intensify, particularly in the northwest, but pose fewer population exposure risks than wet heat waves. The findings highlight that climate change is reshaping heat wave patterns, with humidity playing a critical role in amplifying health risks. Reducing greenhouse gas emissions could significantly lower these risks, emphasizing the importance of global climate action.

1. Introduction

Heat waves, as one of the most severe meteorological disasters, present substantial risks to natural ecosystems, socioeconomic development, and human health (Smoyer-Tomic et al., 2003). In the context of global warming, both the frequency and severity of heat waves have risen substantially worldwide since the mid-20th century, a trend that is projected to continue (Lee et al., 2021; Meehl & Tebaldi, 2004; S. Perkins et al., 2012). The alarming mortality rate seen during heat waves that were considered one-in-100-year events in the climate of 2000 could become commonplace every 10–20 years under the climate conditions of the 2020s (Lüthi et al., 2023). Regionally, a similar intensification of heat waves has been observed in China, where record-breaking heat wave events become increasingly common (Sun et al., 2014; J. Wang & Yan, 2021; Wei, Han, et al., 2023). For instance, the summer of 2022 witnessed the most severe heat wave since 1961, lasting 79 days and affecting over 5 million square kilometers across eastern China (Jiang et al., 2023). This was followed by two record-breaking heat waves in June 2023 and 2024 in northern China, where daily maximum temperatures broke historical extremes at most monitoring stations (Qian et al., 2024). These extraordinary heat wave events seem to be the new norm in China. Projections from climate models, including those from the Coupled Model Intercomparison

Project Phase 5 and 6 (CMIP5 and CMIP6), project further intensification of heat waves in China under future warming scenarios (Ma & Yuan, 2023; Sun et al., 2018; J. Wang et al., 2020; Wei, Wang, et al., 2023). As a densely populated and rapidly developing country with diverse climate types, China faces substantial societal and economic challenges from heat waves, highlighting the urgent need for heat risk reduction strategies in the face of a changing climate.

Conventionally, heat waves are defined as consecutive days exceeding a specified temperature threshold (Perkins & Alexander, 2013). However, it is equally crucial to consider humidity when assessing heat waves, as it directly influences how the human body exchanges heat with the surrounding environment (Mora et al., 2017; J. Yuan et al., 2020). Depending on the prevailing humidity conditions, heat waves are categorized as dry and wet types, each exhibiting distinct characteristics and impacts. Dry heat waves can exacerbate evaporation, intensify droughts, and increase wildfire risk, posing significant threats to agriculture and ecosystems (X. Yuan et al., 2023). In contrast, wet heat waves hinder the body's ability to dissipate heat, posing a serious threat to human health (Mora et al., 2017; Russo et al., 2017). Past research often relied on temperature-humidity indices, like apparent temperature (Luo & Lau, 2019), thermal comfort index (J. Wang & Yan, 2021), wet-bulb temperature (Yu et al., 2021), and wet-bulb globe temperature (Mohammad & Weng, 2024), to classify dry heat waves (low index values) and wet heat waves (high values). However, these indices may mislabel a heat wave as “wet” during extremely hot but dry conditions, or vice versa. An alternative approach combines temperature and temperature-humidity indices (An & Zuo, 2021; Luo et al., 2022), defining dry heat waves as a high-temperature-and low-temperature-humidity index, and wet heat waves as the opposite. But this approach overlooks scenarios where both metrics surpass critical thresholds. Studies like those of Ding and Ke (2014) propose a clearer two-step method that first identifies heat waves by temperatures and then categorizes them as “dry” or “wet” based on relative humidity (RH) levels. This method ensures complete separation between dry and wet events, though these thresholds vary widely (Ha et al., 2022; Ma et al., 2022; Xu et al., 2021). Despite inconsistencies, research on regional dry and wet heat waves in China remains limited, and most studies focused on differences in atmospheric circulation patterns between the two types of heat waves (An & Zuo, 2021; Luo et al., 2022; Ma et al., 2022). Two studies conducted preliminary nationwide analyses of spatial distributions and historical trends of these compound extremes across China (Ding & Ke, 2014; Xu et al., 2021). Nevertheless, systematic assessments of regional disparities in their characteristics remain absent, and projections regarding future changes are still underexplored. Furthermore, the critical influence of heat wave definition requires comprehensive evaluation.

This study aims to provide a systematic assessment of the characteristics, historical trends, and future projections of dry and wet heat waves in China, with particular emphasis on refining their definitions. Our analysis indicates that the risk of heat waves, which has already increased rapidly in China over the past 60 years, is projected to further increase strongly under higher global warming scenarios. In the future, China will be exposed to a heightened risk of wet heat waves. These findings highlight the urgent need for adaptation to heat extremes. Furthermore, our results suggest that the effective mitigation policies that keep a moderate socioeconomic development path can substantially reduce this risk.

2. Materials and Methods

2.1. Observation Data

Observations of daily mean temperature (T_m), RH, and daily maximum temperature (T_{max}) for 2479 meteorological stations during the extended summer (May–September) of 1961–2022 were used in this study. The data set is provided by the National Meteorological Information Center of China and has been quality controlled (Cheng et al., 2024). To ensure data consistency and representativeness, one missing value per summer per year was permitted. If two or more daily values were missing, that summer was marked as a missing summer. Stations with five or more missing summers were then excluded. This process resulted in the selection of 2002 stations. Among them, 19.8% of stations have no missing data and 29.3% only have one missing summer in the 62-year period. To address the potential limitations posed by missing data for heat wave definition, single-day missing values were interpolated with the average of the values from the preceding and following days, while missing values lasting two or more days are interpolated with the climatological mean for the period 1961–1990.

2.2. CMIP6 Outputs

Historical simulations (1961–2014) and future projections (2015–2100) of daily T_m , RH, and T_{max} from CMIP6 were also utilized (Eyring et al., 2016). The projections consist of two scenarios (SSP2-4.5 and SSP5-8.5), which represent combinations of shared socioeconomic pathways (SSPs) (O'Neill et al., 2016) and radiative forcing levels from the representative concentration pathways (RCPs). The SSP2-4.5 (SSP5-8.5) reflects a moderate socioeconomic (high energy-intensive) development path (O'Neill et al., 2016, 2017). Data from 21 CMIP6 models providing outputs for all three variables (T_m , T_{max} , and RH) were utilized (Table S1 in Supporting Information S1). For models with multiple ensemble simulations, the first ensemble member (r1i1p1f1) was selected. To ensure consistency, all model outputs, with different spatial resolutions, were bilinearly interpolated to a $1^\circ \times 1^\circ$ grid. Following the Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report (Lee et al., 2021), 1995–2014 was defined as the current climate period and taken as the baseline for model projections. The period 2081–2100 was defined as the end of the 21st century.

2.3. Population Data

The gridded population count for historical periods and two SSPs (SSP2 and SSP5) was obtained from the NASA Socio-economic Data and Applications Center (Doxsey-Whitfield et al., 2015; Jones & O'Neill, 2016). Historical data, spanning 5 years (2000, 2005, 2010, 2015, and 2020), are available at a horizontal resolution of $1.0^\circ \times 1.0^\circ$. Population projections under the SSPs are provided at 10-year intervals from 2010 to 2100, with a horizontal resolution of $0.125^\circ \times 0.125^\circ$. These projections are upscaled to a $1.0^\circ \times 1.0^\circ$ spatial resolution to match the resolution of the other data sets. This study used the average population for 2000 and 2010 and 2090 and 2100 to represent the current climate period and the end of the century, respectively.

2.4. Identification of Heat Waves

Heat waves are typically defined as periods of consecutive days (i.e., duration) with temperatures exceeding a specific threshold, determined by temperature or related variables. Given the limitations of temperature-humidity indices in defining heat waves, this study employs the two-step classification method. Specifically, a heat wave was identified when either daily T_m or T_{max} exceeded a predefined threshold for at least three consecutive days. T_{max} identifies peak daytime extremes aligned with historical research conventions, while T_m reflects sustained thermal exposure, capturing both prolonged moderate heat and acute extremes. Due to the diverse climate types of China, the 90th percentile of daily temperatures varies dramatically among stations, with T_m ranging from 6.9°C on the Qinghai-Tibet Plateau to 34.0°C in southern China, and T_{max} varying between 14.4°C and 41.8°C (Figure S1 in Supporting Information S1). Localized thresholds that focus on temperature anomalies may yield heat wave events with lower temperatures (Cheng et al., 2024; You et al., 2016). Our approach adopts unified national thresholds to enable cross-regional comparisons and identify nationwide patterns, ensuring consistency across diverse climatic zones. To ensure sufficient heat wave events, we set modest thresholds at 28.4°C for T_m and 34.1°C for T_{max} , respectively, corresponding to the 90th percentile across all meteorological stations in China during the baseline period (1961–1990).

2.5. Methods

Heat waves were characterized by frequency, duration, and intensity. At each station or grid point, frequency was defined as the total number of heat wave days per summer (days), while duration represented the average length of individual heat wave events (days per event). The average wet-bulb temperature (T_w) during a heat wave was used to represent the event's intensity ($^\circ\text{C}$), considering its close relationship to the human body's core temperature. Daily T_w was calculated from daily T_m and RH using the following formula (Stull, 2011):

$$T_w = T_m \times \left[A(RH + B)^{\frac{1}{2}} \right] + \text{atan}(T_m + RH) - \text{atan}(RH - C) + D(RH)^{\frac{3}{2}} \times \text{atan}(E \times RH) - F \quad (1)$$

where A , B , C , D , E , and F are constants ($A = 0.152$, $B = 8.314$, $C = 1.676$, $D = 0.004$, $E = 0.023$, and $F = 4.686$), obtained by fitting functions (Stull, 2011). Physiological observations and models show that the critical T_w for human survivability limit was relatively constant at 33 – 34°C in humid environments but progressively decreased

in hotter, dry ambient environments (Vanos et al., 2023; Vecellio et al., 2023). A $T_{\max}$ value of 46.8°C and a T_w value of 33.3°C were introduced as the danger thresholds for dry and wet heat waves, respectively, at which older adults would suffer physiological heat stroke after 6 hr of exposure under shaded/indoor conditions (Vanos et al., 2023). Population exposure to heat waves was calculated by multiplying the heat wave frequency by the number of exposed people (Ullah et al., 2023). Following Cheng et al. (2024), this study focused on five subregions within China: North China (112–123°E, 34–41°N), the Yangtze River Valley (YRV; 110–123°E, 26–33°N), south China (105–120°E, 18–25°N), southwest China (103–109°E, 27–33°N), and Xinjiang (73–91°E, 35–50°N).

The contributions of T_m , $T_{\max}$, and RH to changes in heat waves were calculated using the climatological average of variables for all years, which means the signal of corresponding variable change was removed (Jones et al., 2015). On this basis, the difference between the original heat wave series and that using controlled variables was used to estimate the contribution from T_m , $T_{\max}$, or RH. Regression analysis and correlation analysis were used to calculate linear trends and linear correlation coefficients. The statistical significance of trends and correlation coefficients was determined using a two-tailed Student's t test.

3. Results

3.1. Identification of Dry and Wet Heat Waves

Figure 1a illustrates the relationship between elevated daily T_m and $T_{\max}$, exceeding their respective thresholds, during heat wave events across China. Notably, the frequency of heat waves peaks near the threshold level, diminishing as temperatures increase beyond them. Consequently, 64.5% of China's heat wave events are characterized by concurrently high T_m and $T_{\max}$, while 19.9% occur with exclusively high T_m and 15.5% with solely high $T_{\max}$. The maximum recorded daily values reached 33°C for T_m and 43°C for $T_{\max}$ under the two categories with solely high metrics, respectively. The narrower range of daily T_m associated with the higher frequency of heat waves suggests that high T_m values typically coincide with high $T_{\max}$ values, although instances of high $T_{\max}$ with lower T_m are possible. In Xinjiang, nearly all heat waves are marked by high $T_{\max}$, occurring most frequently when $T_{\max}$ exceeds its threshold despite daily T_m remaining below its threshold (Figure 1b). This suggests that $T_{\max}$ is a suitable indicator for describing heat waves in Xinjiang. Similarly, $T_{\max}$ appears more crucial than T_m in driving heat wave occurrence in North China and southwest China, whereas T_m plays a more prominent role in the YRV and south China.

Given the positive correlation between RH and temperature, heat waves occurring near the T_m and $T_{\max}$ thresholds, representing the most frequent events, are typically characterized by RH values above 60%, accounting for 80% of all heat wave days (Figure 1c). Correspondingly, heat waves caused exclusively by high T_m are characterized by a higher mean RH (77.0%) than those caused solely by high $T_{\max}$ (63.4%). Considerable regional variability in RH during heat waves is evident. For instance, 98.0% of heat waves in Xinjiang occur with RH below 60%, clustering primarily around 38%. The probability density functions (PDFs) of RH for the three southern subregions exhibit similar shapes, but south China consistently experiences the highest RH values during heat waves, followed by the YRV and then southwest China. These regional differences in RH during heat waves are most evident in the left tail of their PDFs: only 1.5% of heat waves in south China occur with RH below 60%, compared to 8.7% in the YRV and 20.5% in southwest China. North China, by contrast, exhibits the broadest RH range during heat waves, spanning from 11.8% to 98.2%. Accurately distinguishing dry and wet heat waves requires selecting an appropriate RH threshold, traditionally determined empirically. As shown in Figure 1d, the proportion of heat waves classified as wet decreases as the RH threshold increases. With an RH threshold of 60%, wet heat waves account for 84.0% of all heat waves in China, aligning with the findings of Ding and Ke (2014). Nearly all heat waves in Xinjiang are classified as dry, while most in south China are wet. Wet heat waves also dominate in the other three subregions, accounting for 56.7% in North China, 85.4% in southwest China, and 97.0% in the YRV. However, increasing the RH threshold to 80% substantially decreases the proportion of wet heat waves, with all regions exhibiting proportions below 25%. These findings highlight the sensitivity of heat wave classifications, and consequently the subsequent analyses, to the chosen RH threshold. However, establishing a scientifically robust and universally applicable RH threshold remains challenging. To address this, we adopted a data-driven approach, defining the RH threshold as the average RH across all heat wave days in China from 1961 to 2022 (RH = 69%), facilitating a relative assessment of humidity conditions during heat waves by comparing regional humidity levels to the national average.

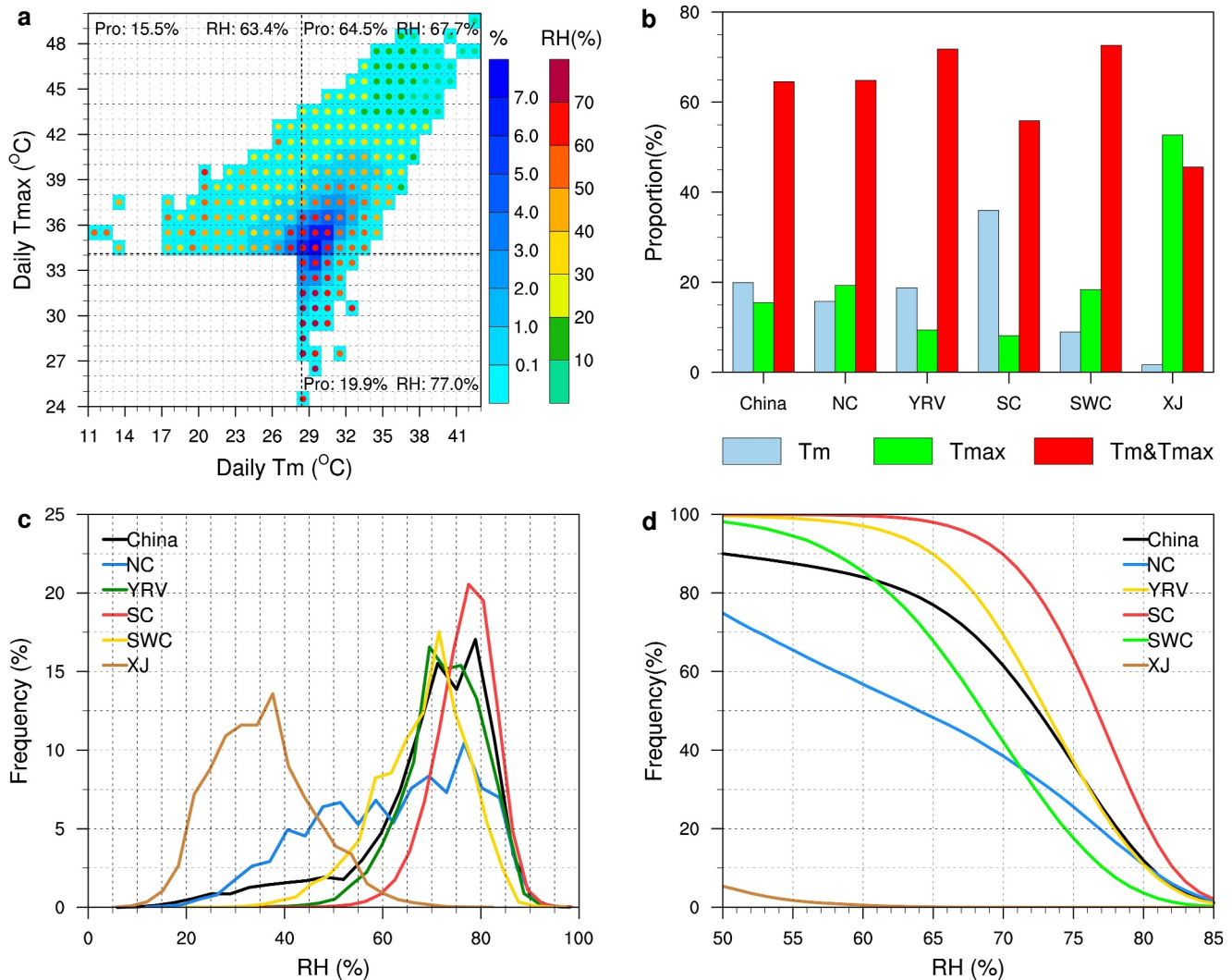


Figure 1. Characteristics of daily mean temperature (T_m), relative humidity (RH), and daily maximum temperature (T_{max}) during heat waves in China. (a) Proportions of heat waves as a function of daily T_m and T_{max} in China. Dots represent the average RH during heat waves under the corresponding configuration. The numbers are cumulative proportions (Pro) and average RH for the corresponding quadrants. (b) Proportions of three heat wave categories with solely high T_m (blue), solely high T_{max} (green), and both high T_m and T_{max} (red). (c) Probability density function of RH during heat waves in China and its five subregions, that is North China (NC), Yangtze River Valley (YRV), south China (SC), southwest China (SWC), and Xinjiang (XJ). (d) Proportions of wet heat waves as a function of RH in China and its five subregions.

3.2. Different Climatic Characteristics of Dry and Wet Heat Waves

Figure 2a illustrates the mean heat wave frequency across China for the period 1961–2022, highlighting that heat waves are most prevalent in northwestern and eastern China. We prioritized five subregions for subsequent in-depth analysis. The mean summer heat wave frequencies across all stations in China, North China, the YRV, south China, southwest China, and Xinjiang are 17.3, 10.4, 34.4, 15.6, and 16.4 days, respectively. Using a RH threshold of 69%, China experiences an average of 6.0 days of dry heat waves per summer, with the highest frequency observed in Xinjiang, 2.7 times the national average (Figures 2b and 2d). As an arid region, Xinjiang's dry climate and intense solar radiation are conducive to high-temperature events (Ding et al., 2011). The frequency of dry heat waves in North China, the YRV, and southwest China exceeds the national average, whereas south China experiences only about half the national average.

Wet heat waves, by contrast, are more frequent than dry heat waves, averaging 11.3 days per summer across China. The highest frequencies are observed in south China and the YRV, at 3.4 and 2.3 times the national average, respectively (Figures 2c and 2d). These regions are characterized as humid regions (Huang et al., 2016).

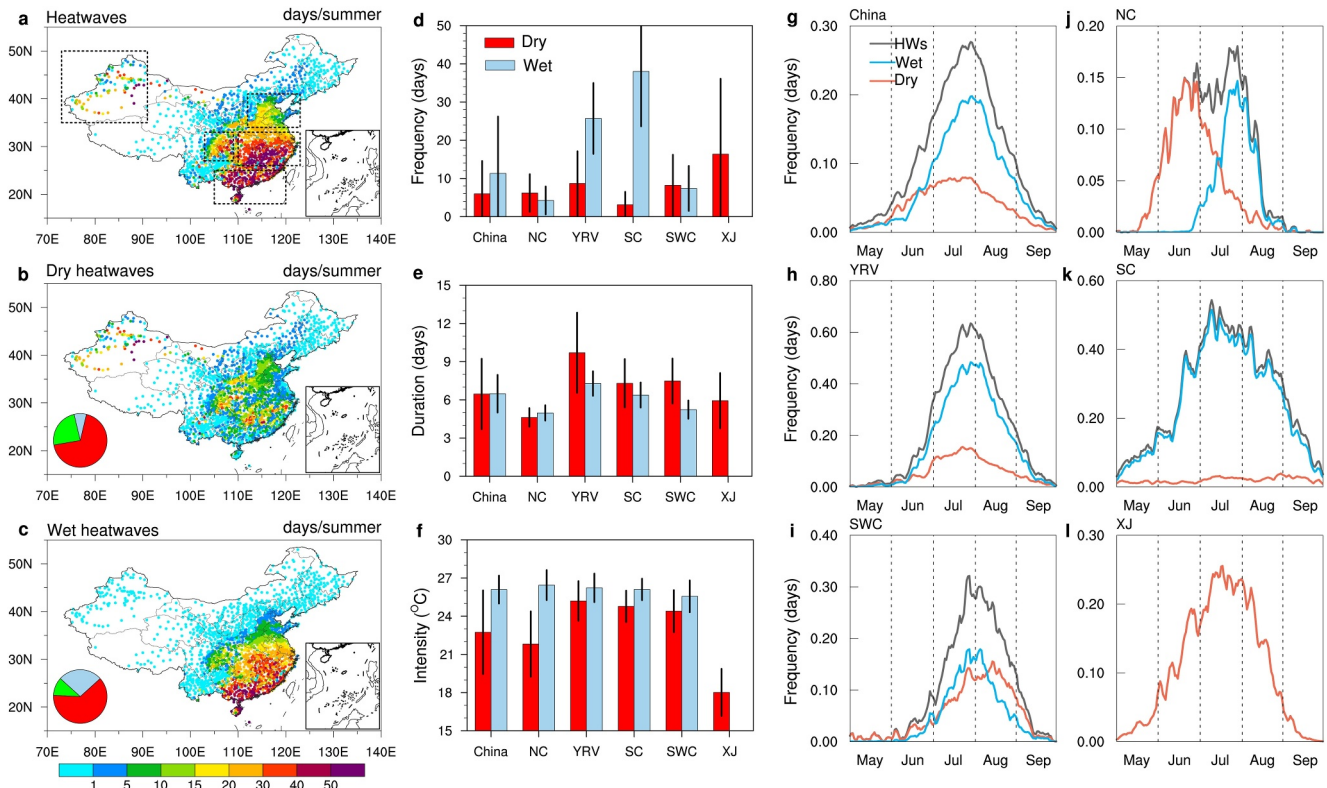


Figure 2. Climatological mean of dry and wet heat waves in China for the period 1961–2022. Spatial distribution of summer mean frequency of (a) total, (b) dry and (c) wet heat waves. Dashed boxes in panel (a) denote the five subregions. Pie charts in panels (b) and (c) represent the proportions of three heat wave categories with solely high T_m (blue), solely high T_{max} (green), and both high T_m and T_{max} (red). Regional mean (d) frequency, (e) duration, and (f) intensity of dry and wet heat waves in China and its five subregions. The perpendicular black lines at the top of the bars represent one standard deviation of regional variables. (g–l) Intraseasonal variations in the regional frequency of total, dry, and wet heat waves in China and its five subregions.

In transitional regions such as North and southwest China, the frequencies of dry heat waves (6.2 and 8.2 days, respectively) and wet heat waves (4.2 and 7.4 days, respectively) are roughly equal. Additionally, the proportion of high T_{max} exceeding the threshold is higher for dry heat waves, while high T_m prevail during wet heat waves. This suggests dry heat waves are more strongly influenced by high T_{max} , while wet heat waves are more closely associated with high T_m . Dry heat waves often occur briefly during the daytime, while wet heat waves tend to be more persistent and potentially pose a greater risk to human health. The spatial distribution patterns of dry and wet heat waves broadly align with the findings of Ding and Ke (2014) but contrast with those of Xu et al. (2021). This divergence primarily stems from the application of localized thresholds, which identifies extensive high-frequency regions of wet heat waves spanning from the Tibetan Plateau to south China (Xu et al., 2021).

Dry and wet heat waves exhibit significant differences in duration, intensity, and associated meteorological characteristics. While the national average duration for both dry and wet heat waves is 6.5 days, substantial regional variability in dry heat wave duration is observed between and within subregions (Figure 2e). In southern China—encompassing the YRV, south China, and southwest China—dry heat waves persist longer than wet ones, whereas wet heat waves exhibit slightly longer durations in North China. The YRV records the longest average heat wave durations, at 9.7 days for dry heat waves and 7.3 days for wet heat waves. Wet heat waves are associated with a narrower range of T_m values, lower T_{max} values, and higher RH values (Figure S2 in Supporting Information S1). Compared to dry heat waves, higher RH and comparable T_m during wet heat waves contribute to significantly higher T_w , that is, heat wave intensity. On average, wet heat waves are 16% more intense than dry ones in China, with average T_w values of 26.1°C and 22.5°C, respectively (Figure 2f). Furthermore, the intensity of wet heat waves exhibits minimal spatial variability. The higher RH values in the three southern subregions contribute to greater dry heat wave intensity than North China and Xinjiang.

Figures 2g–2i present the intraseasonal variations in heat wave frequency across China and its five subregions from 1961 to 2022. Nationally, the mean monthly heat wave frequencies from May to September are 0.6, 2.7, 7.2, 5.6, and 1.2 days, respectively, exhibiting a single peak from late July to early August. The intraseasonal patterns of dry and wet heat waves generally mirror the total heat wave pattern, with dry heat waves peaking in late July and wet heat waves in early August. Ding and Ke (2014) conducted monthly scale analyses of the seasonal cycle, demonstrating patterns consistent with our findings. The YRV and southwest China exhibit similar single-peak intraseasonal patterns. However, the YRV is characterized by a higher proportion of wet heat waves, while southwest China experiences a more balanced distribution of dry and wet heat waves. Both south China and Xinjiang exhibit peak heat wave activity occurring early in July. Despite these temporal similarities, all heat waves in Xinjiang are dry, whereas the majority in south China are wet. Unlike the other subregions, North China exhibits a distinct bimodal intraseasonal pattern of heat wave occurrence, with a peak in late June driven primarily by dry heat waves, and a second peak in late July dominated by wet heat waves. Therefore, North China experiences a shift in the dominant heat wave type throughout the summer, transitioning from dry heat waves in June to predominantly wet heat waves thereafter. This pattern of intraseasonal variability in North China aligns with findings in previous literature (An & Zuo, 2021) and is closely related to the intraseasonal evolution of temperature and RH, with North China being distinctly dry in early summer (average RH below 69%) before transitioning to a wetter regime in midsummer (average RH exceeding 69%) (Figure S3 in Supporting Information S1).

3.3. Historical Changes in Dry and Wet Heat Waves

Figure 3a presents the trends in dry heat wave frequency across China from 1961 to 2022. Notably, nearly all stations display an increasing trend, with 43.3% exhibiting statistically significant changes. This increase is most pronounced in mideastern China, particularly in the YRV (1.5 days/decade) and southwest China (1.3 days/decade). Nationally, after a slight decline until the late 1980s, dry heat wave frequency has risen steadily at a rate of 2.0 days per decade (Figure 3c). For wet heat waves, 32.8% of stations exhibit significant trends, with 84.6% of these trends being positive (Figure 3b). The national increase averages 0.5 days per decade, following a distinct four-phase evolutionary pattern: decreasing (1960–1980s), increasing (1990s), stabilizing (2000s), and surging (post-2000s). In particular, the increase in wet heat waves since the late 2000s (4.5 days per decade) is much more pronounced than dry heat waves (Figure 3c), resulting in an increase in the proportion of wet heat waves (Figure S4 in Supporting Information S1). Despite discrepancies in the reported trends arising from heterogeneous data sources, divergent study periods, and inconsistent heat wave definitions, a robust consensus emerges across studies regarding a marked intensification of heat wave activity since the 1990s, particularly for wet heat waves (Ding & Ke, 2014; Xu et al., 2021). The prevalence of wet heat waves under climate change poses a severe threat to human health. South China has experienced the most pronounced surge, with frequency rising continuously since the 1970s at a rate of 3.9 days per decade—7.2 times the national average increase (Figure S4 in Supporting Information S1). The YRV and southwest China exhibit a mix of increasing and decreasing trends in wet heat wave frequency at individual stations, resulting in a relatively stable regional average. In North China, both dry and wet heat waves exhibit comparable changes and similar long-term trends, with a decline until the 1990s, followed by an increase accompanied by interdecadal fluctuations over the subsequent three decades (Figure S4 in Supporting Information S1).

The increase in dry heat waves is primarily attributed to decreasing RH during summer, especially in the YRV and southwest China (Figure 3g), where notable reductions in RH were observed (Figure S5c in Supporting Information S1). Wet heat wave increases are primarily linked to temperature rises (Figure 3h). As the climate warms, more days exhibit high T_m and T_{max} , increasing wet heat wave frequencies (Figure 3h and Figure S5d in Supporting Information S1). South China experiences the steepest temperature-driven (both T_{max} and T_m) increase, with T_{max} changes driving wet heat waves in southwest China, and T_m changes influencing other regions (Figure 3h). Notably, North China exhibits a rise in the proportion of wet heat waves and a significant increase in RH during dry heat waves, suggesting a transition of heat waves toward wet types (Figure S5e in Supporting Information S1).

Figures 3e and 3f reveal that dry heat waves have more pronounced increases in both duration and intensity than wet ones across China. Notable increases in duration are observed for dry heat waves in North China and Xinjiang, as well as for wet heat waves in south China and North China, driven by rising T_m and T_{max} (Figure 3e). Conversely, the duration of both dry and wet heat waves in the YRV has decreased, though these trends lack

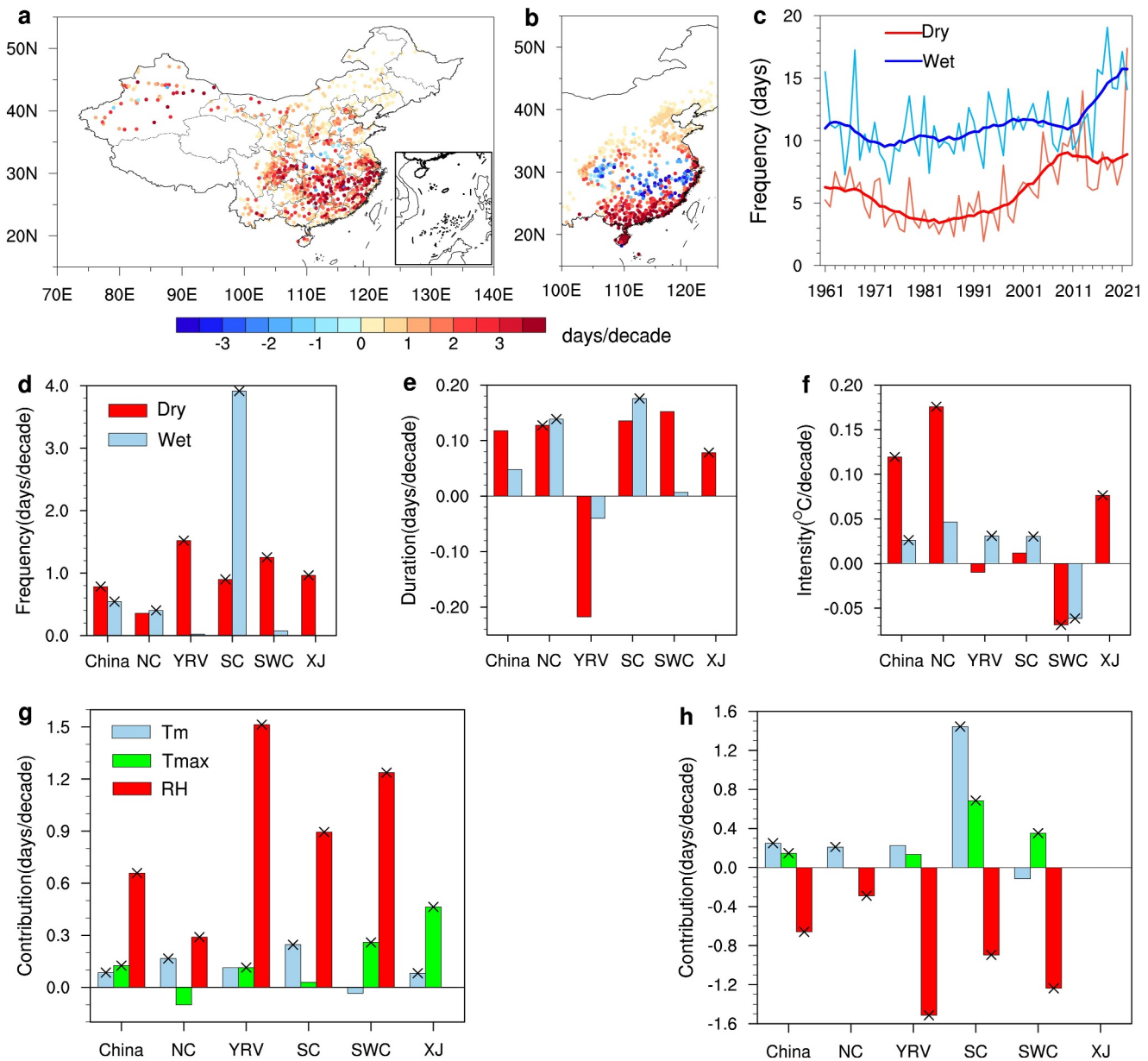


Figure 3. Historical changes in dry and wet heat waves for the period 1961–2022. Trends in frequency of (a) dry and (b) wet heat waves. Only stations with significant trend at the 0.1 level are shown. Linear trends of (c) and (d) frequency, (e) duration, and (f) intensity of dry and wet heat waves in China and its five subregions. Contributions of T_m , T_{max} , and RH to the changes in the frequency of (g) dry and (h) wet heat waves. The cross in panels (d–h) indicates a statistically significant change.

statistical significance. This more pronounced warming, coupled with increasing RH, contributes to a greater increase in the intensity of dry heat waves than wet ones (Figure 3f and Figures S5a and S5c in Supporting Information S1). Regionally, North China and Xinjiang have seen significant intensification of dry heat waves, with North China's changes attributed to a synergy of increased T_m and RH, whereas Xinjiang's are driven solely by warming. In the YRV and south China, despite reduced RH, warming has led to a significant intensification of wet heat waves. Overall, heat waves across China have become more intense and persistent, with marked changes in North China linked to higher T_m and RH and reduced T_{max} . Conversely, in southwest China, increasing T_{max} alongside decreasing T_m and RH decrease heat wave intensity.

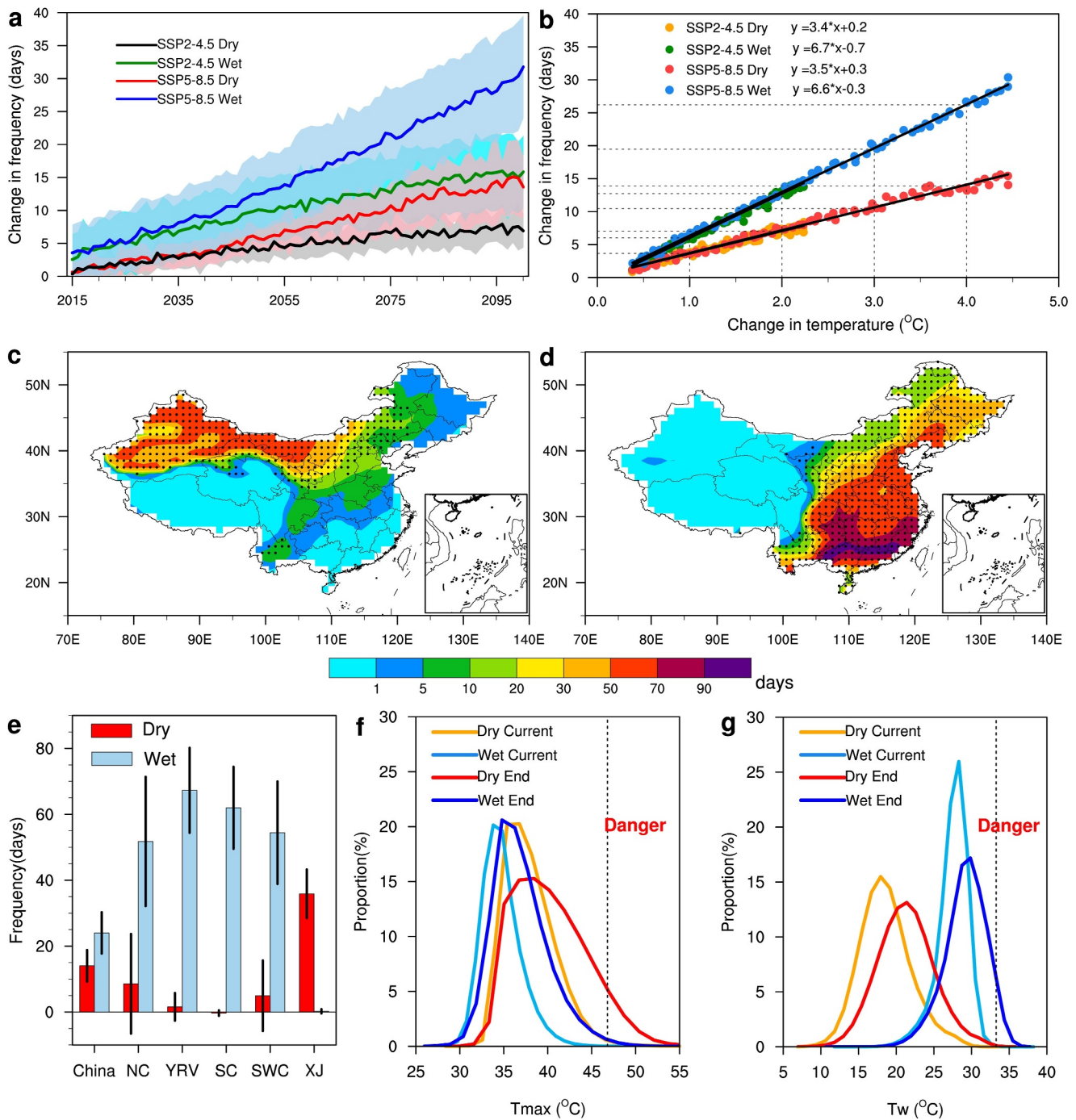


Figure 4. Future changes in dry and wet heat waves relative to the current climate period (1995–2014). (a) Time series of changes in the frequency of dry and wet heat waves from 2015 to 2100 under the SSP2-4.5 and SSP5-8.5 scenarios. Shading represents one standard deviation. (b) Changes in the frequency of dry and wet heat waves as a function of the annual mean temperature increase. Changes in the frequency of (c) dry and (d) wet heat waves by the end of the century under the SSP5-8.5 scenario. Dotted areas indicate agreement among more than 90% (18 of 21) of the model projections on the sign of change (positive or negative). (e) Average changes in frequency of heat waves over China and its subregions. Perpendicular black lines at the top of the bars represent one standard deviation. Probability density functions of (f) T_{max} and (g) wet-bulb temperature (T_w) during heat waves in China during the current climate period and the end of the century under the SSP5-8.5 scenario. Vertical lines represent the thresholds of dangerous heat waves.

3.4. Future Projections of Dry and Wet Heat Waves

Figure 4a illustrates projected changes in the frequency of dry and wet heat waves across China, from 2015 to 2100 under the SSP2-4.5 and SSP5-8.5 scenarios. CMIP6 models exhibit considerable variability in climatological values of T_m , T_{max} , and RH; therefore, model-specific thresholds are employed (Figure S6 in Supporting Information S1). The ensemble mean projects a more substantial increase in wet heat wave frequency than dry heat waves. Under the SSP2-4.5 scenario, dry heat waves are projected to increase by 0.8 days per decade, and wet heat waves by 1.5 days per decade, with trends significant at the 0.1 level. The SSP5-8.5 scenario projects even larger increases, with dry heat wave frequency projected to increase by 1.7 days per decade and wet heat wave frequency by 3.3 days per decade. By the end of the century (2081–2100), dry and wet heat wave frequencies are projected to surge by $140.7\% \pm 48.3\%$ (one standard deviation; the same below) and $224.1\% \pm 58.7\%$ under the SSP5-8.5 scenario, respectively, compared with the current climate period (1995–2014), closely linked to global temperature increase (Figure 4b). Note that the heat wave increases by about the same magnitude as the average temperature rises in the SSP2-4.5 and SSP5-8.5 scenarios. For every 1°C increase in global mean temperature, the frequency of dry (wet) heat waves in China is projected to increase by about 3.4 days (6.6 days). The SSP5-8.5 scenario projects a global temperature increase of $\sim 4^\circ\text{C}$ by 2100, resulting in a 28.2-day increase for wet heat waves, doubling that of dry heat waves (14.1 days).

Regional-specific projections indicate the arid region in northwestern China will experience the most substantial increase in dry heat waves, with an increment of 35.9 ± 7.4 days in Xinjiang by the end of the century under the SSP5-8.5 scenario (Figure 4c). Eastern China's humid regions will see a substantial increase in wet heat waves (Figure 4d). By the end of the century under the SSP5-8.5 scenario, wet heat wave frequency could increase by 67.3 ± 12.9 in the YRV, 62.0 ± 12.5 in south China, and 54.4 ± 15.6 days in southwest China (Figure 4e). Notably, North China is projected to experience more substantial increases in wet heat wave frequency, 6 times that of dry heat waves. Besides, heat waves in China are anticipated to become more intense and prolonged under future warming, especially for wet ones. By the end of the century, SSP5-8.5 projects an average duration increase of 6.9 ± 2.0 days for dry heat waves and 9.6 ± 6.5 days for wet heat waves (Figure S7b in Supporting Information S1). Intensities of dry and wet heat waves in China will increase by $2.3 \pm 0.8^\circ\text{C}$ and $1.6 \pm 0.5^\circ\text{C}$, respectively (Figure S7d in Supporting Information S1). Consequently, 7.9% of dry heat waves and 5.4% of wet heat waves are projected to be deadly dangerous, with T_{max} exceeding 46.8°C for dry heat waves and T_w exceeding 33.3°C for wet heat waves (Figures 4f and 4g).

Due to demographic trends under both SSP2 and SSP5, population exposure to heat waves in China will increase until midcentury and then decline (Figure S8 in Supporting Information S1). Despite larger projected population declines under SSP5, exposure to heat waves remains higher than under SSP2 throughout the 21st century. Similar to the changes in heat wave frequency, population exposure to wet heat waves is projected to increase more remarkably and robustly, with a value of $100.7 \pm 35.8\%$ by the end of the century under SSP5-8.5, doubling that of dry heat waves. By then, 68.6 million and 265.4 million people, accounting for 10.1% and 39.1% of the total, respectively, will be exposed to dangerous dry and wet heat waves (Figure 5d), with over 3.3 million and 154.7 million enduring more than 10 days of such conditions annually. Regionally, North China sees the highest exposure increase to heat waves, particularly for wet heat waves with end-century SSP5-8.5 projections showing a $162.1 \pm 74.5\%$ rise (Figures 5a–5c). Though North China also witnesses a 73.4% increase in dual exposure to dry heat waves, it is not as robust as the wet heat waves.

Humans are more vulnerable to deadly dangerous wet heat waves than to dry heat waves. Under the SSP5–8.5 scenario, population exposure to such extreme wet heat waves is projected to reach 6.3 million person-days by the end of the century (Figure S9 in Supporting Information S1). A moderate socioeconomic development path (SSP2-4.5) can significantly mitigate the risk of heat waves, aligning spatial patterns but with smaller increases in heat wave frequency and exposure (Figure S10 in Supporting Information S1). In particular, the proportions of extreme dry and wet heat waves by the end of the century in the SSP2-4.5 scenario are only 2.5% and 0.3%, respectively. The heat wave frequency and population exposure to extreme heat waves decline drastically, with virtually no risk increasing across China. Thus, climate adaptation policies can effectively reduce the risk of heat waves in China.

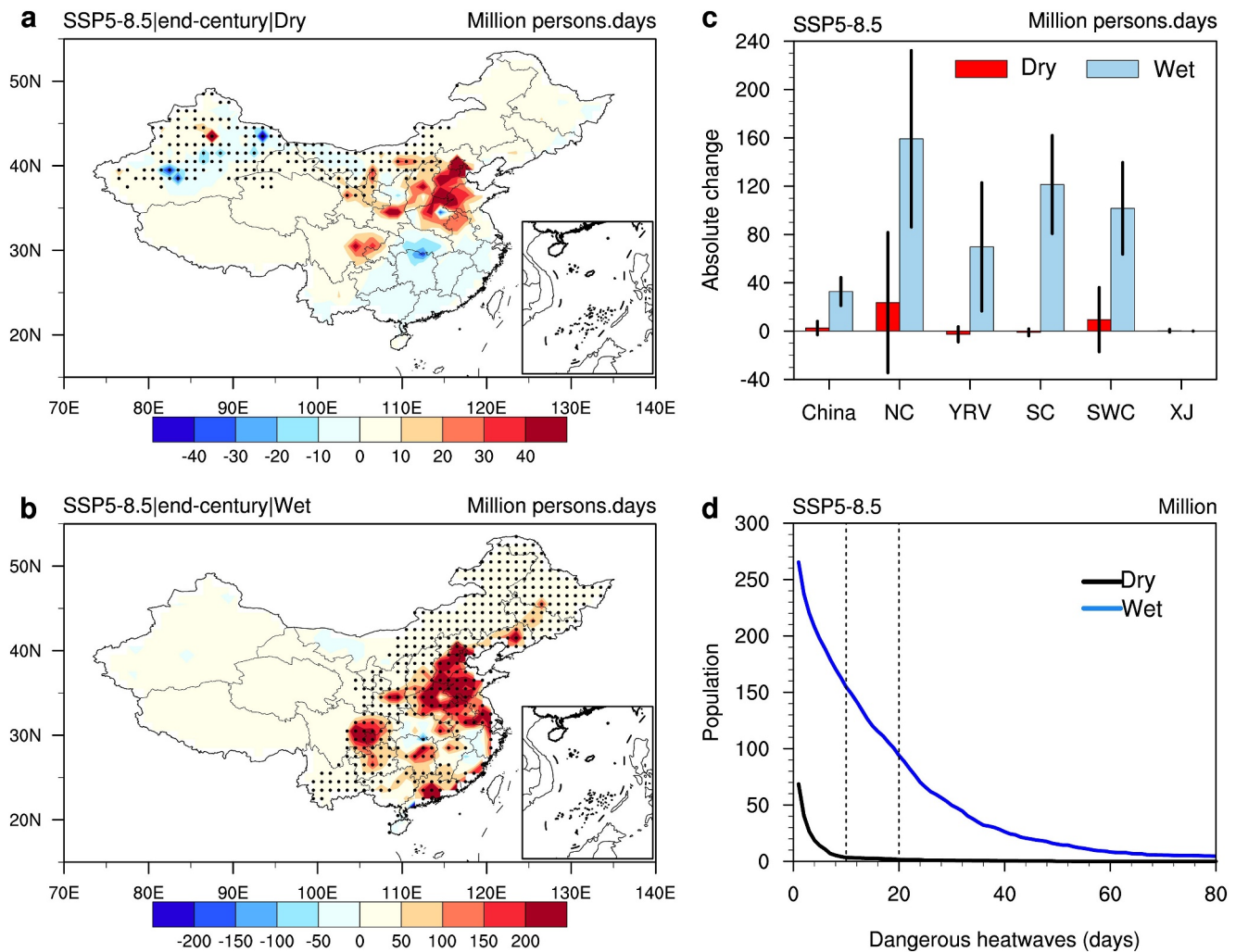


Figure 5. Future changes in population exposure to dry and wet heat waves relative to the current climate period (1995–2014). Changes in population exposure to (a) dry heat waves and (b) wet heat waves by the end of the century under the SSP5-8.5 scenario. Dotted areas indicate agreement among more than 90% (18 of 21) of the model projections on the sign of change (positive or negative). (c) Average changes in population exposure to heat waves over China and its subregions. Perpendicular black lines at the top of the bars represent one standard deviation. (d) Population count as a function of dangerous heat wave days in China by the end of the century under the SSP5-8.5 scenario.

4. Conclusions and Discussions

Understanding the spatial distribution and temporal evolution of heat waves under a warming climate is crucial for developing effective climate change adaptation strategies. This study distinguished dry and wet heat waves using a data-driven RH threshold and investigated the historical characteristics and future changes utilizing observational data and CMIP6 projections. Dry heat waves are predominant in arid northwestern China, whereas wet heat waves are more prevalent in humid southern China. Transitional regions, including North China and southwest China, experience comparable frequencies of both dry and wet heat waves. Nationally, the duration of wet heat waves in China is comparable to that of dry heat waves, but the intensity of wet heat waves is 17.9% higher. North China uniquely exhibits a bimodal seasonal pattern of heat wave occurrence, with dry heat waves peaking in early summer and wet heat waves in midsummer. Over recent decades, both dry and wet heat waves have become more frequent, with the most substantial increases in dry heat waves observed in mideastern China and wet heat waves in south China. The increase in dry heat waves is mainly attributed to decreasing RH, whereas wet heat wave trends are predominantly associated with rising temperatures. Furthermore, heat waves in China have become more persistent, with North China exhibiting the most significant increase in dry heat wave duration

and south China experiencing the greatest increase in wet heat wave duration. Increasing T_m and RH values have resulted in a more pronounced intensification of dry heat waves, particularly in North China. Consideration of both dry and wet heat waves reveals that North China has faced the most significant intensification in terms of both intensity and duration.

Projections indicate a more substantial increase in wet heat wave frequency in China than dry heat waves throughout the 21st century, with increases of $224.1\% \pm 58.7\%$ and $140.7\% \pm 48.3\%$, respectively, by the end of the century relative to the current climate period under SSP5-8.5. Wet heat waves substantially increase across eastern China, and dry heat waves surge in northwestern China. The increase in population exposure to wet heat waves in China is projected to nearly double that of dry heat waves, with North China facing the most severe risk. More frequent and intensified heat waves could lead to 10.1% and 39.1% of the population experiencing dangerous dry and wet heat waves, respectively. Furthermore, China has implemented a more liberal policy to increase the birth rate nowadays, and the population might be underestimated at the end of this century, implying a more serious demographic risk from heat waves (X. Yuan et al., 2019).

Establishing a robust and universally applicable definition is crucial for studying compound heat wave events influenced by humidity and temperature. This study adopts a two-step classification method that fully separates dry and wet events, and unified thresholds that enable consistent cross-regional comparisons. Despite being statistically derived, the RH threshold aligns closely with a physiologically informed threshold (66% at 36°C environmental temperature), beyond which heat stress becomes uncompensable for healthy adults (Vecellio et al., 2023). We recognize that unified thresholds may overlook variations in heat tolerance across age groups and fitness levels (Cvijanovic et al., 2023). However, maintaining this approach remains necessary to identify nationwide patterns and facilitate regional comparisons. Future studies should develop tailored thresholds for vulnerable populations to refine heat wave definitions. Addressing this challenge requires enhanced collaboration between health professionals and climate scientists, informed by comprehensive evidence on the differential impacts of dry and wet heat waves on human health.

Alterations in atmospheric moisture substantially contribute to heat wave intensification. Dry heat waves are often linked to high-pressure systems featuring descending air motions, reduced moisture, and increased solar radiation (Cai et al., 2024; Deng et al., 2019; P. Wang et al., 2019). In contrast, the atmospheric conditions associated with wet heat waves are more complex, with varying atmospheric circulation patterns regionally (Ha et al., 2022; Luo et al., 2022). Projected regional differences between dry and wet heat waves highlight the critical role of humidity, suggesting future research should focus on the mechanisms driving the observed and projected changes, including the contributions of various natural and anthropogenic factors.

Understanding the uncertainties of climate models should be another noteworthy issue. All 21 models generally replicate the observed increasing frequencies of both dry and wet heat waves in China over recent decades (Figure S11 in Supporting Information S1). Despite intermodel variations in the warming magnitude, the projected surge in heat wave frequency remains sufficiently robust to withstand these discrepancies. China has experienced a substantial increase in wet heat waves over recent decades and would be further enhanced in a warmer future, particularly in the densely populated southern humid regions. This projected increase in wet heat waves and associated risks is theoretically supported by an increase in atmospheric water-holding capacity. To mitigate these risks, it is crucial to implement appropriate adaptation measures, formulate effective policies, and make proactive, forward-looking arrangements. For example, early warning systems could integrate humidity thresholds to better reflect physiological risks, particularly in regions like North China where wet heat wave exposure is projected to surge. Urban infrastructure in high-risk areas may require heat-resilient designs to alleviate compounding effects of temperature and humidity. These recommendations aim to bridge climate science with societal needs, while detailed policy implementation necessitates interdisciplinary expertise beyond the meteorological scope.

Data Availability Statement

The observation data are available via <http://data.cma.cn> (Cheng et al., 2024). The outputs of CMIP6 could be freely obtained from <https://esgf-node.llnl.gov/search/cmip6> (Eyring et al., 2016). The population data are available at <https://sedac.ciesin.columbia.edu/data/collection/popdynamics/sets/browse> (Doxsey-Whitfield et al., 2015).

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